Zhengyan Huan

PhD Candidate in Electrical and Computer Engineering at Tufts University

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Education

Tufts University — Medford, MA
PhD in Electrical and Computer Engineering — *September 2024–Expected May 2029*

- Research interests: generative models, constrained generation and inference, and reinforcement learning.

University of Pennsylvania — Philadelphia, PA
M.S. in Electrical Engineering — *September 2022–May 2024*

Hefei University of Technology — Hefei, China
B.Eng. in Electronic Information Engineering — *September 2018–June 2022*

Research Experience

Tufts University — Medford, MA
Graduate Research Assistant, advised by Prof. Shuchin Aeron — *September 2024–Present*

- Conduct research on generative models, inverse problems, and reinforcement learning, with an emphasis on constrained generation and inference; authored three first-author papers, including one accepted and two under review.

Hefei University of Technology — Hefei, China
Undergraduate Research Assistant, advised by Prof. Bin Chen — *September 2020–June 2022*

- Investigated modulation-format optimization for optical communication systems, resulting in two published papers (one first-authored) and a granted patent.

Selected Research Projects

Primal-Dual Flow Matching for Sample-Wise Constrained Generation
First author; submitted to NeurIPS 2026 — May 2025–May 2026

- Developed Primal-Dual Flow Matching (PDFM), a framework that directly controls a user-specified constraint-satisfaction rate using only binary membership-oracle feedback, without requiring differentiable distances, projections, or convex constraint sets.
- Established theoretical guarantees on distributional fidelity and constraint satisfaction under flow matching approximation and optimization errors, and derived a deterministic-policy-gradient-inspired surrogate for training with non-differentiable constraints.
- Increased physical-consistency rates for generated Kuramoto–Sivashinsky dynamics from 14.19% to 97.74% on clean data and from 9.82% to 90.48% on degraded data while maintaining competitive distributional fidelity.

Constraint-Aware Flow Matching via Randomized Exploration
First author; published in TMLR, April 2026 — November 2024–September 2025

- Developed constraint-aware flow matching methods for constraints specified by either differentiable distances or binary membership oracles, using randomized velocity exploration to learn a constraint-aware mean flow for the oracle-only setting.
- Designed a two-stage training strategy that confines randomized exploration to later flow segments, reducing computational cost while preserving the trade-off between constraint satisfaction and distributional fidelity.
- Demonstrated applicability to non-convex, disconnected, and empty-interior constraints; reduced constraint violations from 9.14% to 1.12% for brightness-controlled MNIST generation and generated adversarial examples against hard-label black-box classifiers.

The Ensemble Inverse Problem: Applications and Methods
First author; under review at TMLR — January 2025–January 2026

- Formulated the Ensemble Inverse Problem, which recovers posteriors associated with unknown priors from sets of observations without explicit or iterative use of the shared forward model at inference time.
- Developed ensemble inverse flow matching and diffusion models that condition posterior sampling on both individual observations and a learned permutation-invariant representation of the observation ensemble, enabling generalization to unseen priors without handcrafted summary statistics.

- Validated the framework on inverse imaging, particle-physics unfolding, and seismic full-waveform inversion; on an unseen $t\bar{t}$ process, reduced Wasserstein distance from 3.36 to 0.23 for transverse momentum and from 3.29 to 0.66 for energy relative to conditional flow matching.

Policy Gradient and Actor-Critic Approaches to Quickest Change Detection
First author; published in IEEE TSP, June 2026 September 2023–January 2025

- Designed a trajectory-level probabilistic policy that makes decisions until the first alarm and deterministically keeps alarming thereafter, treating the first alarm as the stopping time instead of optimizing independent per-step labels.
- Derived one policy-gradient and two actor-critic algorithms that optimize a QCD-specific cumulative reward without known pre- or post-change distributions; achieved near-CUSUM synthetic performance and competitive or superior performance across three real-world detection tasks.

Technical Skills

Programming: Python, MATLAB
Machine Learning: PyTorch, JAX, TensorFlow

Teaching Experience

Tufts University Medford, MA
Teaching Assistant *Fall 2025–Spring 2026*

- Supported High-Dimensional Probability (Fall 2025) and Machine Learning via Sequential Interaction (Spring 2026) by helping design homework assignments and exams, grading coursework, and holding office hours.

Selected Peer-Reviewed Publications

Zhengyan Huan, Jacob Boerma, Li-Ping Liu, and Shuchin Aeron. *Constraint-Aware Flow Matching via Randomized Exploration, Transactions on Machine Learning Research (TMLR), 2026.*

Zhengyan Huan, Xiaodong Wang, and Xiao-Yang Liu. *Policy Gradient and Actor-Critic Approaches to Quickest Change Detection, IEEE Transactions on Signal Processing, 2026.*

Dongsheng Ding, **Zhengyan Huan**, and Alejandro Ribeiro. *Resilient Constrained Reinforcement Learning, AISTATS, 2024.*

Camila Pazos, Shuchin Aeron, Pierre-Hugues Beauchemin, Vincent Croft, **Zhengyan Huan**, Martin Klassen, and Taritree Wongjirad. *Towards Universal Unfolding of Detector Effects in High-Energy Physics Using Denoising Diffusion Probabilistic Models, SciPost Physics Core, 8(4):064, 2025.*